Development of a biologically inspired artificial lateral line for undulatory robotics

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Background

- Fish-inspired undulatory flexible robots offer an advantage over traditional underwater vehicles in complex environments.
- These vehicles must be able to detect and react to fluid flow rate.
- Sensors traditionally used to measure fluid flow rate cannot be effectively integrated on flexible vehicles.
- Fish have a lateral line system that allows them to sense the flow of the water around them, which could be replicated for underwater vehicles.

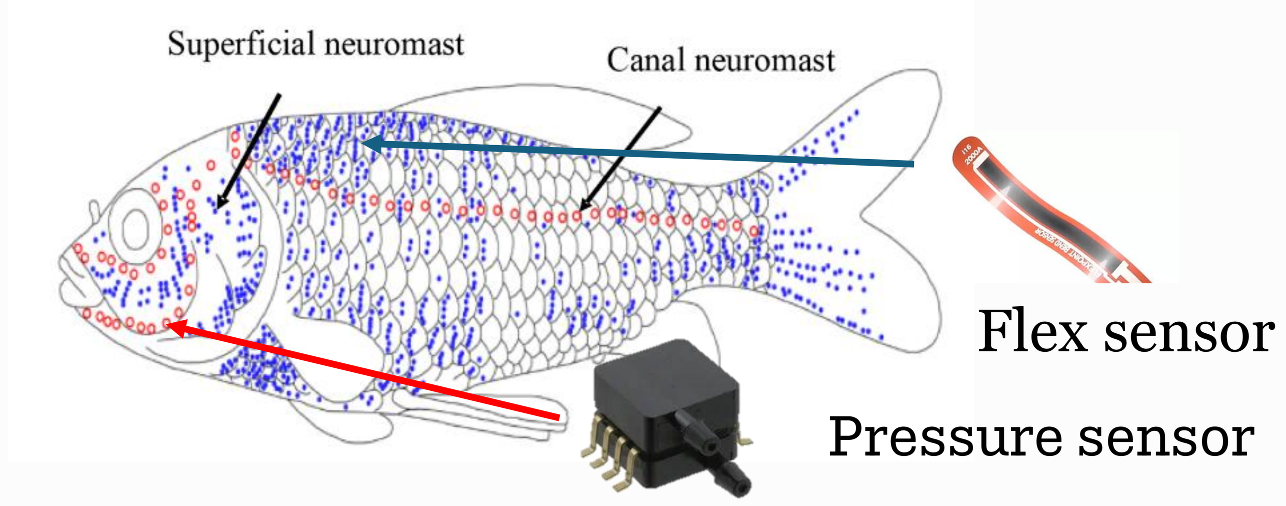

Fig. 1. Physical sensors to replace biological lateral line structures [1]

Objective

Replicate the biological lateral line system using commercial flex sensors and differential pressure sensors

Materials & Methods

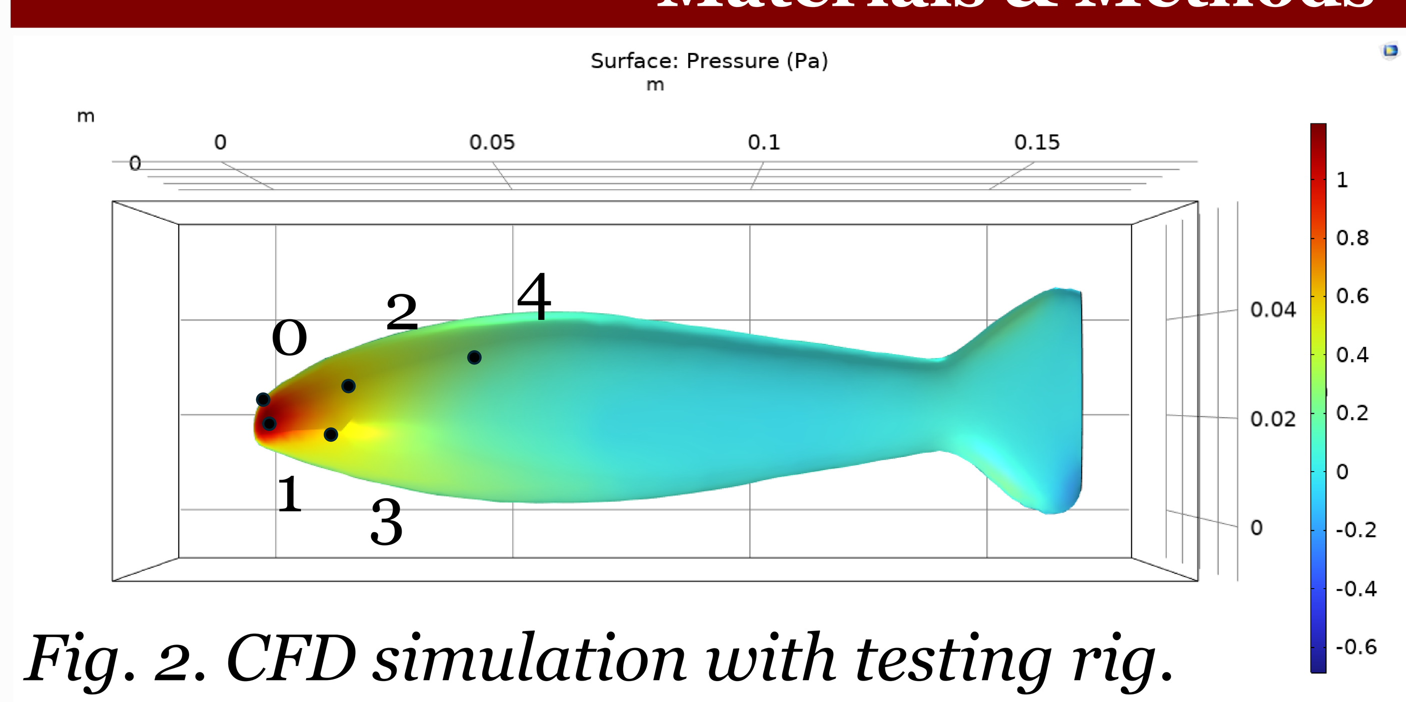

Fig. 2. CFD simulation with testing rig. Final sensor locations are marked.

Design:

Created a trout-inspired rig to house sensors in small-scale tests. Featured ports for sensors and an external electronic system.

Preliminary testing:

Collected data from 5 differential pressure sensors in the water tunnel at angles between 15 and -15 degrees in 2.5 degree intervals at 3 different speeds determined by swimming speeds experienced by undulatory swimmers such as trout.

Fig. 3. Rig in water tunnel

Results

The differential pressure sensors were found to be sensitive enough to detect slight changes in pressure as flow conditions changed.

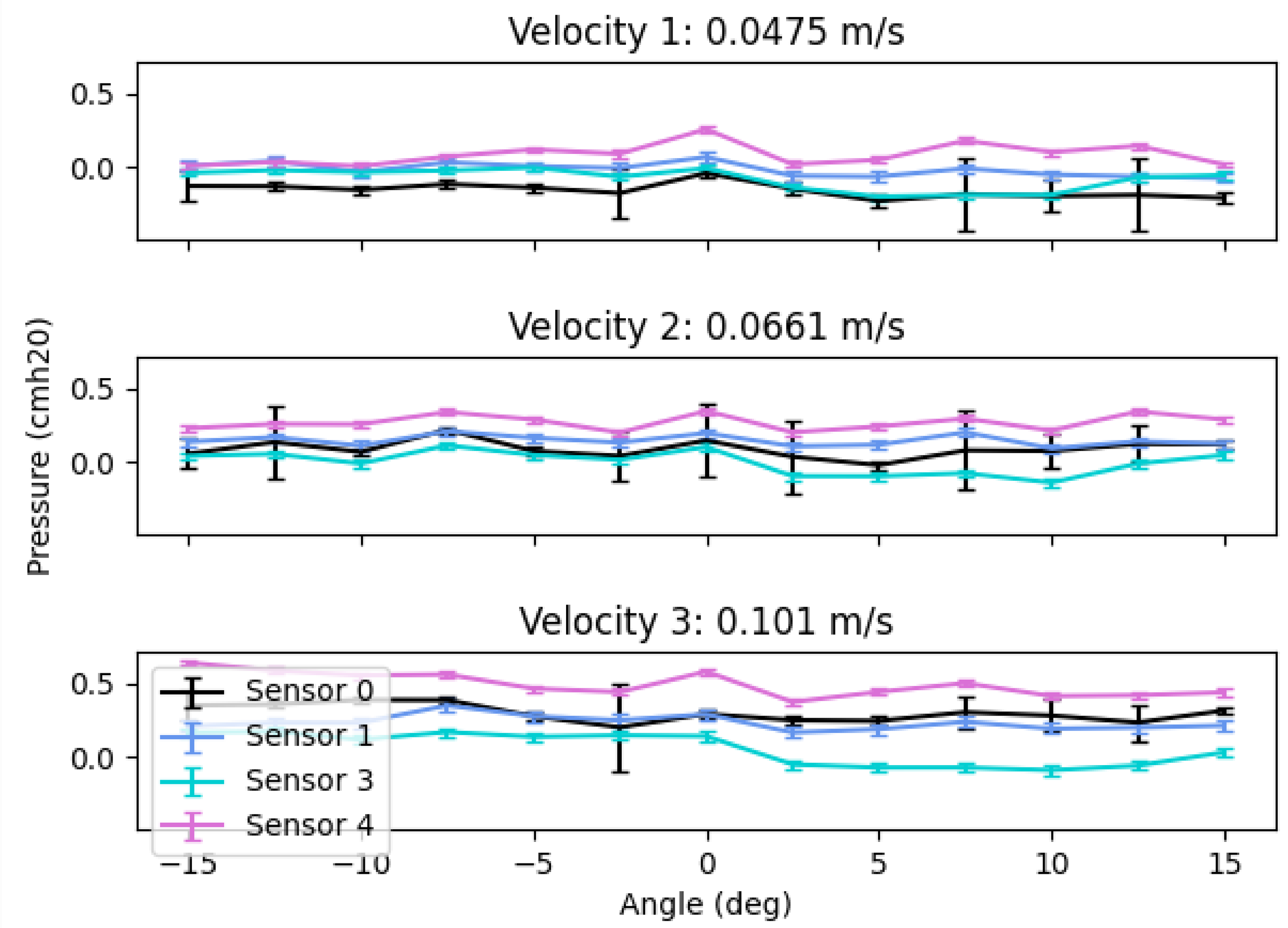

Fig. 4. The pressure measured by each sensor vs horizontal angle of attack for each of the 3 velocities samples. The pressure was averaged at each condition. The data collected at 0 m/s was used as a baseline and subtracted from each datapoint to highlight small changes in pressure. Sensor 2 was omitted due to measurement issues.

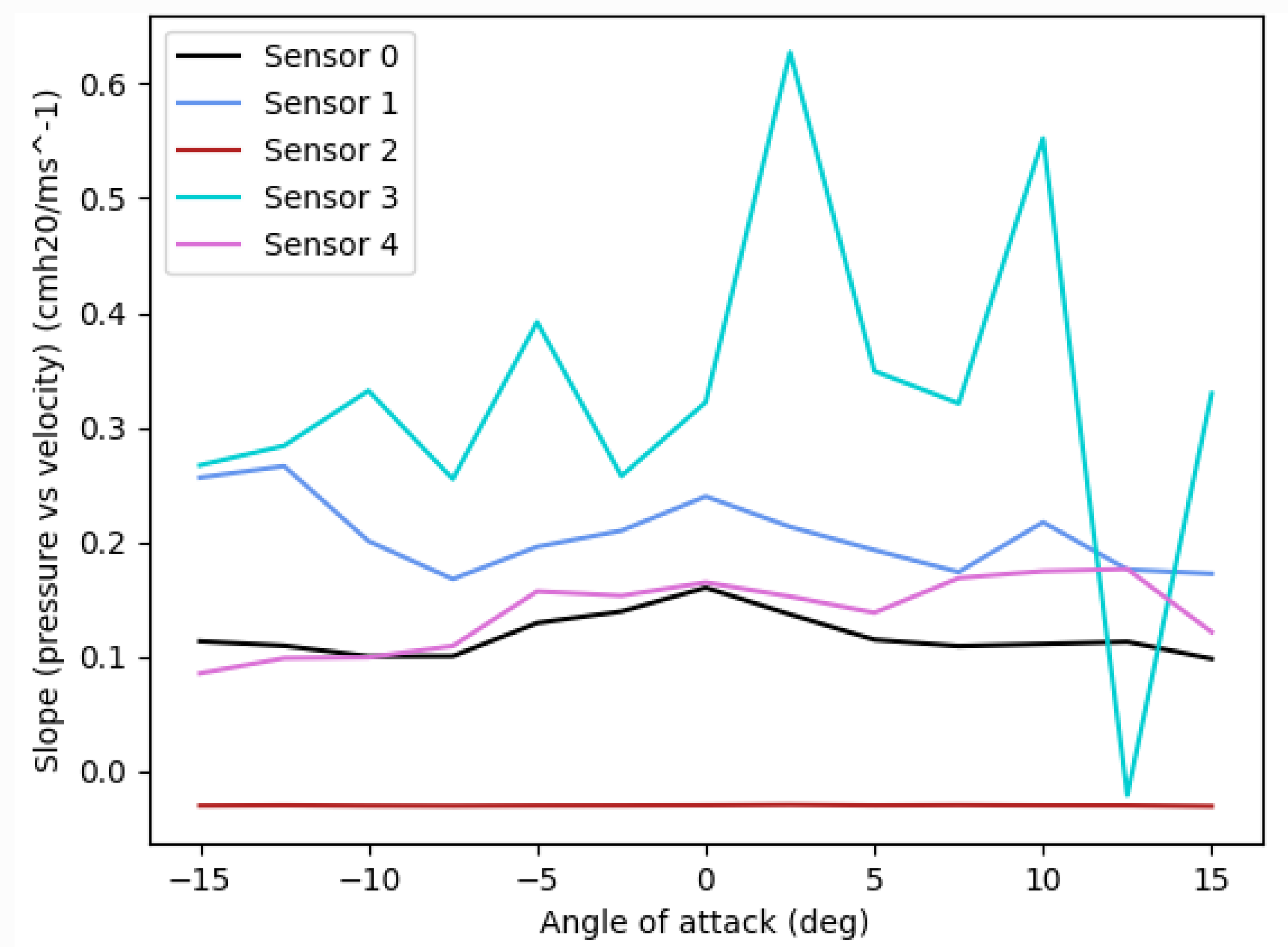

Fig. 5. Slope derived from pressure and velocity data graphed vs the horizontal angle of attack. Sensor 3 and sensor 1 were most sensitive to changes in pressure as velocity changed.

From the preliminary testing, the data shows patterns that would be detectable by a machine learning model. The combined data from multiple sensors could be used to train a model to predict flow field parameters based on sensor feedback. This would improve the controllability of undulatory robotics.

Future Directions

Test with flex sensors in addition to pressure sensors.

Use a machine learning model to predict the flow parameters based on sensor data.

Test under undulatory conditions in the water tunnel.

Implement on undulatory bio-mimetic robot.

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